

o you 2m



9.1 – 9.3 – Waves + Diffraction + Interference

Example 1:

Zoe is trying to look at her new super cool laser under a microscope... She notices that as the light goes from the classroom air (assume vacuum conditions) through the microscope slide, it bends. Tara is VERY accurate at eyeballing angles. She is certain that the light hit the plastic slide at an angle of incidence of 30 degrees. Everybody knows that plastic microscope slides have an index of refraction of 1.51.

- What is the velocity of the laser as it passes through the microscope slide?
- What angle does it refract to? (Feel free to ask Willebrord Snell).
- Zoe accurately counts the frequency of the laser while in air to be 5.45×10^{14} Hz. What colour light was the laser?
- What is the period?
- Is it possible for the laser to totally internally reflect in this situation? Explain.

Example 2:

Sarah L is at the local pond teaching swimming/lifeguarding to her friends. Emma and Yuki are practicing throwing lifeguard floaties at Sarah. They both throw their floaties at the same time but they hit in the air and miss Sarah. Both floaties happen to land in the water at the exact same time just 2m apart from each other. They happen to cause ripples in phase with one another and cause an interference pattern.

- Sarah is treading water exactly on the third nodal line, 7.5 m away from where Emma's floatie landed and 6.5m from Yuki's floatie. What is the wavelength that these ripples have?
- This happened yesterday as well, but yesterday Sarah was on the second nodal line just 4m away from Emma's floatie. Assuming the wavelengths were the same, how far away was the Sarah from Yuki's floatie?

- Why wouldn't the equation

$$\frac{x_n}{L} = \frac{\left(n - \frac{1}{2}\right)\lambda}{d}$$

be used here?



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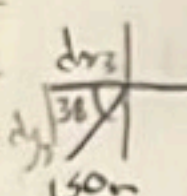
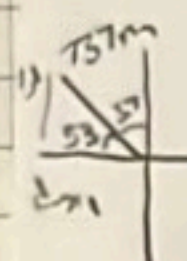
Name:

.3 Quiz V3

SOH CAH TOA

Thinking (8 marks)

Clearly answer the following questions showing all your work. Round your final answers to 3 sd.



x comp:

$$\cos 53 = \frac{dx_1}{737m}$$

$$dx_1 = \cos 53 \cdot 737m$$

$$dx_1 = 444m [W]$$

$$\cos 68 = \frac{dx_2}{412m}$$

$$dx_2 = \cos 68 \cdot 412m$$

$$dx_2 = 154m [E]$$

$$\cos 38 = \frac{dx_3}{150m}$$

$$dx_3 = \cos 38 \cdot 150m$$

$$dx_3 = 118m [W]$$

$$dx_T = dx_1 + dx_2 + dx_3$$

$$dx_T = 408m [W]$$

y comp:

$$\sin 53 = \frac{dy_1}{737m}$$

$$dy_1 = \sin 53 \cdot 737m$$

$$dy_1 = 589m [N]$$

$$\sin 68 = \frac{dy_2}{412m}$$

$$dy_2 = \sin 68 \cdot 412m$$

$$dy_2 = 382m [S]$$

$$\sin 38 = \frac{dy_3}{150m}$$

$$dy_3 = \sin 38 \cdot 150m$$

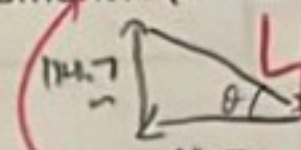
$$dy_3 = 92.3m [S]$$

$$dy_T = dy_1 + dy_2 + dy_3$$

$$dy_T = 114.7m [N]$$

$$\therefore d_T =$$

$$424m [N 16^\circ N]$$



$$C^2 = a^2 + b^2$$

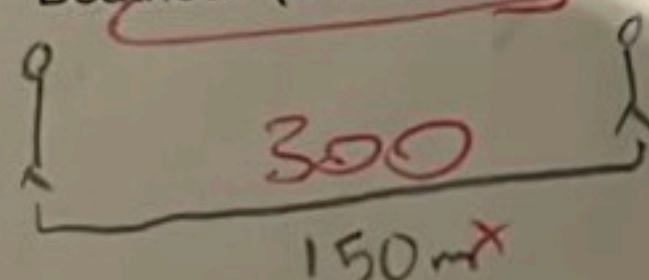
$$C = \sqrt{(114.7m)^2 + (408m)^2}$$

$$C = 424m$$

1. Diba walked all the way from home to school, covering a total of 737 m [N 37° W]. She then realized it was Sunday and the Bills played in a few minutes! Because she didn't want to miss the game she sprinted 412 m [22° E of S] to Brandon's, but he didn't answer the door. So she ran another 150 m [W 38 S] to Sara's where she could finally catch the end of the game. Phew! What was Diba's total displacement? (4 marks)

1. Manuela and Beatrice haven't seen each other in over 12 hours! They are both approaching STL when they spot each! They are each 150 m away from STL (from different directions). Manuela had stopped moving because she was so excited, but then starts to accelerate at 1.7 m/s² towards Beatrice. Whereas Beatrice breaks into a constant sprint of 3.8 m/s. How far does Manuela travel before she finally meets up with Beatrice? (4 marks)

$$12h \times \frac{60min}{1h} \times \frac{60s}{1min} = 43200s$$



B:

$$v = 3.8m/s$$

$$\Delta v = \frac{d}{t}$$

$$d = v \cdot t$$

$$d = 3.8m/s \cdot t$$

M:

$$a = 1.7m/s^2$$

$$v_i = 0m/s$$

$$d = v_i \cdot t + \frac{1}{2} a t^2$$

$$d = \frac{1}{2} (1.7m/s^2) t^2$$

$$d = 0.85m/s^2 t^2$$

$$d = 0.85m/s^2 (11.2s)^2$$

$$d = 106.6m$$

$$150m = 3.8m/s \cdot t + 0.85m/s^2 t^2$$

$$0 = 0.85m/s^2 t^2 + 3.8m/s \cdot t - 150m$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(3.8m/s) \pm \sqrt{(3.8m/s)^2 - 4(0.85m/s^2)(-150m)}}{2(0.85m/s^2)}$$

$$x = 11.2s \rightarrow \text{when they collide}$$

$\therefore$ Manuela travels 107m before she meets up with Beatrice



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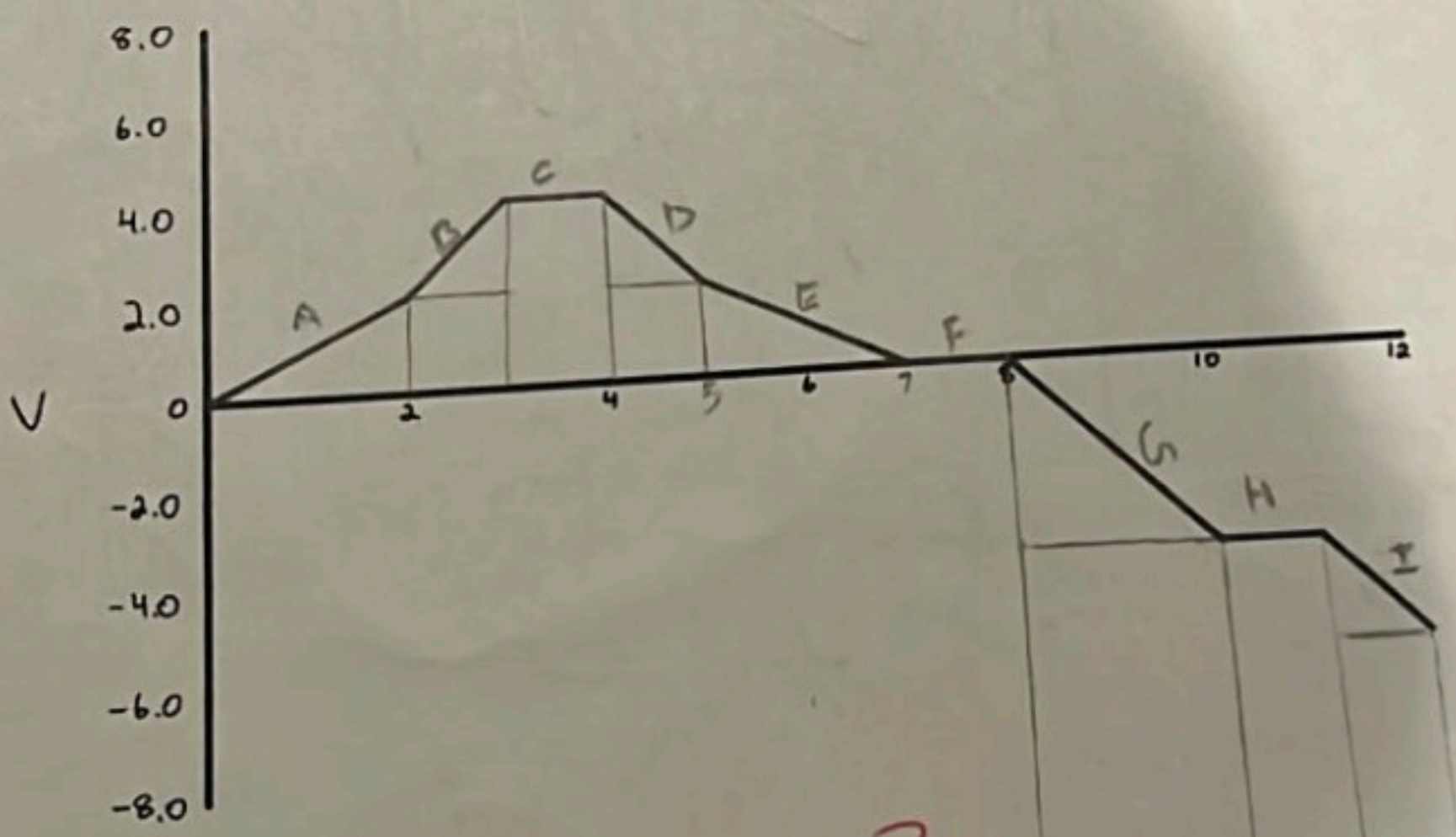
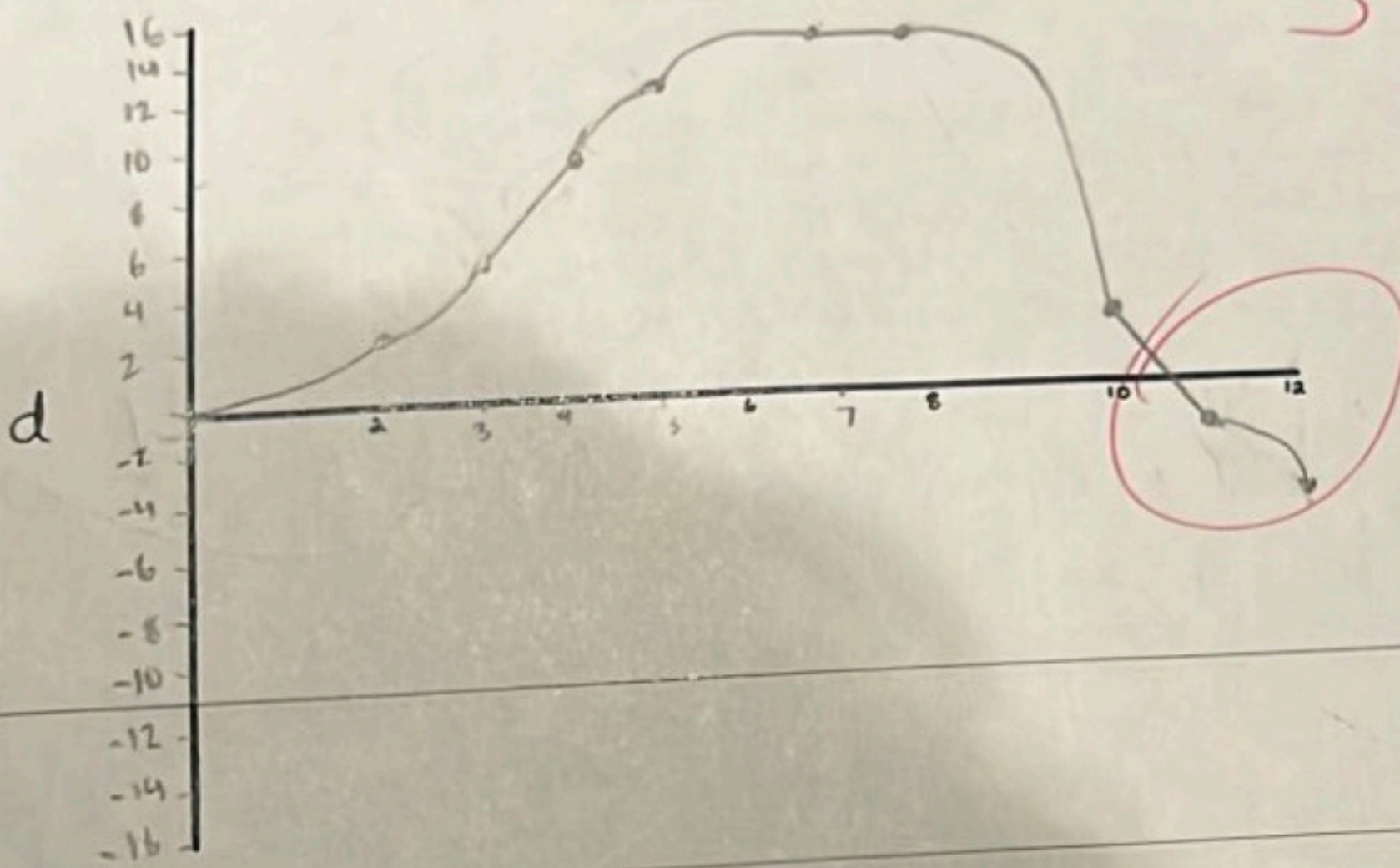
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A: $\frac{2-2}{2} = 2m$ B: $\frac{1-2}{2} = 1m$ C: $\frac{2-1}{2} = 4m$ D: $\frac{2-1}{2} = 3m$ E: $\frac{2-2}{2} = 2m$ F: $\frac{2-2}{2} = 0m$ G: $\frac{2-(-4)}{2} = -4m$ H: $\frac{2-(-4)}{2} = -4m$ I: $\frac{1-(-2)}{2} = -3m$

Communication (7 marks)

1. Convert the velocity time graph to a displacement time graph. (4 marks)



$\uparrow = +$
 $\downarrow = -$
 $\uparrow = +$

- Answer the following questions (3 marks):
- At what times was acceleration the greatest? (magnitude) 2-3s, 4-5s, 8-10s, 11-12s
 - What was the displacement at 11s? -2.00m
 - What was the acceleration from 8-10s? 2.00ms⁻² [downwards direction]



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